Exercise 1.4.1

Task 1

Suppose you take out a loan with a principal amount of #100,000. If the annual interest rate on the loan is 5% and you plan to repay it in 3 years, how much interest will you owe at the end of the loan term?

For this task write a function called `interest()` that accepts the 3 parameters as input arguments and calculates the amount of interest owed at the end of the loan term. The 3 input arguments should be in the given order:

- The principle loan amount
- The annual interest rate
- The number of years in which the loan should be repayed

Once you have written the function, call it with the provided parameters:

- Principal (P) = #100,000
- Rate (R) = 5% per annum
- Time (T) = 3 years

Solution

```
# Function to calculate simple interest
def interest(principle, rate, years):
    interest = (principle * rate *years) /100
    #print(interest)
    return interest

    # Given values
    p = 100000
    R = 5
    T = 3
    print(interest(P, R, T))
```

```
### Run your function with the provided parameters here:
```

Task 2

Write a function called `celsius_to_fahrenheit()` that converts a temperature from degrees Celsius to degrees Fahrenheit.

You can use this formula:

$T_c * 1.8 + 32$

where T_c is the temperature in degrees celsius to be converted.

Once you have written the function, use it to convert the following temperatures from Celsius to Fahrenheit:

1. 100
2. 75
3. 120

Solution

```
# Function to convert celsius to fahrenheit
def celsius_to_fahrenheit(celsius):
    fahrenheit = (celsius *1.8) + 32
    return fahrenheit

    # convert the given temperatures
    print("100 C=", celsius_to_fahrenhit(100), "F")
    print("75C =", celsuis_to_fahrenthit(75), "F")
    print("120C =", celsius_to_fahrenment(120), "F")
```

```
### Run your function with the provided parameters here:
```